

PLC Implementation in GX Works2

Adaptive Demand Allocation Architecture (ADA) — Programming Reference

Ladder Logic, Tag Nomenclature, and an Experimental Structured Text (ST) Translation

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Date July 17, 2026
Status Programming Reference — For Review

This document is a programming reference companion to the *Thermofluid Analysis of Architecture II* and its PLC Implementation section: nomenclature, a consolidated tag/ID lookup, every ladder logic block in one place, and an experimental Structured Text translation flagged accordingly. Design rationale, derivations, and open items live in the parent architecture document, not here.

1 Nomenclature

1.1 Instruction Set (FX-Series-Style, via GX Works2)

Table 1: Ladder instruction abbreviations used throughout this document.

Instruction	Meaning
LD / LDI	Load contact (normally-open / normally-closed) – starts a rung
AND / ANI	Series AND (normally-open / normally-closed)
OR / ORI	Parallel OR (normally-open / normally-closed)
OUT	Drive a coil/flag
SET / RST	Latch ON / latch OFF (retains state until explicitly reset)
MOV	Move/copy a value into a register
MUL / DIV / ADD / SUB	16-bit arithmetic
DMUL / DADD / DDIV	32-bit (double-word) arithmetic – required when a value exceeds the 16-bit range
INC	Increment a register by 1
CMP	Compare, sets a 3-bit result (>, =, <)
PLS	One-scan pulse on a rising edge
FROM / TO	Read from / write to an intelligent module's buffer memory
K	Prefix for a constant (decimal)

1.2 Device Prefixes

Table 2: PLC device types.

Prefix	Device type
X	Physical digital input
Y	Physical digital output
M	Internal relay/flag (not a physical point)
D	Data register (16-bit unless part of a 32-bit pair)
T	Timer
AI-#	Analog input channel (placeholder label, pending real tag numbers)

1.3 Process and Architecture Terms

Table 3: ADA-specific abbreviations.

Term	Meaning
ADA	Adaptive Demand Allocation Architecture
P1 / P2 / P3	Priority tiers: T ₅ trim-heat / PO tank maintenance / T ₁₀ palm oil melt
LSM / HSM / FSM	Trunk flow states: Low/High/Full Supply Mode – 1, 2, or 3 destinations served simultaneously
PO tank	Palm olein tank – whichever of TA/TB/TC is currently designated
S_{casual} , $S_{\text{immediate}}$	Tank scoring functions (volume-weighted / temperature-weighted)
SV	Solenoid valve (water routing)
PBV	Pneumatic ball valve (oil dispense)
HE1–3	Auxiliary immersion heating elements, 12 kW each
AODD	Air-operated double-diaphragm pump (T ₁₀ → T ₅ transfer)
VFD-01 / VFD-02	Trunk circulation drive / oil distribution drive
T_f	Dynamic upper thermal boundary (T ₅ safety cutoff)
Eco/Efficiency/Standard/Full Throttle	P3 operating-posture modes, driven by T ₅ 's level (α_S)
E_{total}	Sum of remaining energy demand across all currently- served branches ($E_\alpha + E_5 + E_{C^*}$), Section 3.11
P_{base} , E_{req}	E_{total} converted to an average power over the planning window, and the throttled power request derived from it (Dynamic Power Evaluation, Section 3.13)
Planning window	5 min (proposed, not user-confirmed) – the time basis used to convert remaining energy to an average power figure; see the warn-box in Section 3.11

2 Tag ID Tables

2.1 Digital I/O (Confirmed)

Table 4: Digital inputs.

Tag	Correspondence	Notes
X0	E-STOP	NC, hardwired safety assumed upstream
X1	Automation Enable/Disable	
X2	Circulation Pump (manual)	
X3	Oil Distribution Pump (manual)	
X4	Jacket Pump (manual)	Unused – Y4 runs continuously, see Section 3.4 no
X5	AODD Pump (manual)	
X6	Float – HEA, NC	Warning only
X7	Float – SOL, NC	Warning only
X10	Loop Exit	Manual-mode only
X_Confirm	PO tank designation confirm	Momentary, HMI or push-button
X_EStopReset	E-STOP reset	Momentary, must follow physical E-STOP clear
X_AlarmReset	Master alarm acknowledge	Momentary, HMI or push-button

X_Confirm, X_EStopReset, and X_AlarmReset are used by name in the ladder logic but have not been assigned real numeric addresses – add them here once confirmed, keeping the naming consistent with X0–X10 above.

Table 5: Digital outputs.

Tag	Correspondence	Notes
Y0	E-STOPPED lamp	
Y1	Automation Enabled lamp	
Y2	Circulation Pump – HSM	
Y3	Oil Distribution Pump active	
Y4	Jacket Pump active	Always on with M2_SystemOK – no manual/auto split
Y5	AODD Pump active	
Y6	Circulation Pump – LSM	
Y7	Heating Element 1	12 kW, now gated on M2_SystemOK (Section 3.14)
Y10	Heating Element 2	12 kW, gated on M2_SystemOK
Y11–Y14	SV100–SV103	T5/TA/TB/TC branch valves
Y15/Y16	SV104/SV105	Solar selector: through/bypass
Y17	Circulation Pump – FSM	
Y20	Heating Element 3	12 kW, gated on M2_SystemOK
Y21–Y24	PBV100–PBV103	T5/TA/TB/TC oil valves
Y25	Solar Cycle Active lamp	

Y4 (Jacket Pump active) intentionally does not follow the Manual/Automatic Gating Pattern used by the other pumps – it runs continuously whenever the system is not E-STOPped, with no manual/auto split. X4 is therefore intentionally unused.

2.2 Analog Inputs (Channel Count Confirmed, Tags Pending)

Table 6: Analog inputs.

Channel	Correspondence	Register
AI-1–4	Level: T5/TA/TB/TC (4-20mA ultrasonic)	$\alpha_S, \alpha_{10A,B,C}$
AI-5–8	Temp: T5/TA/TB/TC (4-20mA Pt100)	$\theta_S, \theta_{10A,B,C}$
AI-9/10	Temp: Solar loop / Thermal battery	$\theta_{\text{Sol}}, \theta_H$
AI-11	Flow: Trunk/outlet (4-20mA, confirmed)	Flow rate

2.3 Data Registers (D) by Function

Table 7: Register groups – see the full architecture document for exact per-register purpose within each group.

Group	Range/registers
AI scaling	D0, D2, D4 (per-channel, repeated pattern)
PO designation	D50 (pending), D51 (active)
T_{water}	D200–D204, D_Ptotal, D_Twater_target
Mode	D_Mode, D_AlphaS
Suspicion	D_Suspicion
E_{total} (now computed, not just consumed)	D700–D744, D_EalphaTA/TB/TC, D_E5, D_E5sensible, D_E5loss, D_EC_star, D_Etotal – Section 3.11
Dynamic power	D_PbaseW, D_Pbase, D_Twater, D_TempDiff, D_TempCalc, D_Ereq_kW – Section 3.13
Aux staging	D_AuxStage, D_Ereq_kW
Scoring	D410–D418, D_ScoreCasualXX, D_ScoreImmXX, D_EligCasualXX, D_MaxScore, D_MaxID, D_RankedTankTemp
Mass conversion	D510, D520, D_MassT5/TA/TB/TC, D_HeightT5/TA/TB/TC
T_f	D330–D371, D_ThetaH, D_ThetaS, D_Mp (now sourced , Section 3.12), D_Tf
Arbiter	D_NeedCount, D_ServeCount
Fault detection	D600–D622, D_DeltaThetaXX, D_MaxHeatRateXX, D_MaxCoolRateXX, D_DeltaMassXX

D_Mp is now resolved. It is computed, not entered or measured directly: T5's remaining capacity to reach 100% full, derived entirely from T5's own live mass (Section 3.12). This was an open item in the previous version of this document and is now closed.

2.4 Working Flags (M) by Function

Table 8: Flag groups.

Group	Flags
Safety/mode	M1_ESTOP, M2_SystemOK, M3_AutoEnabled
PO designation Demand	M0–M3 M_P1Demand, M_P2Demand, M_D_T5/TA/TB/TC
Arbiter	M_Serve_T5/TA/TB/TC, M_Challenger
T_f /P1	M_TfInterlock, M4_ThrottleOrDivert, M_BatchRelease
P2	M_Overrun_Pulse, M_PromptActive, M_HMIPrompt, M_Alarm, M_DisableFeed
Oil distribution	M_DemandX/Y/Z, M_ServeX/Y/Z
Fault detection	M_FaultHeatingXX, M_FaultCoolingXX, M_FaultLevelXX, M_HighLimitXX, M_TAAActiveTransfer
Fast override (proposed, T5 only, see Section 3.13)	M_FastOverride_T5

2.5 Timers (T)

Table 9: Timers.

Timer	Purpose
T0	10s refresh oscillator (self-resetting)
T5	Suspicion meter 5-min no-response auto-increment

Every address above is an illustrative placeholder, coordinated only to avoid collisions with itself – confirm a real, complete address map before programming.

3 Ladder Logic – Full Listing

3.1 First-Scan Initialization

```
LD    M8002
RST   M_Alarm
RST   M_HighLimitTA
RST   M_HighLimitTB
RST   M_HighLimitTC
RST   M_FaultHeatingTA  RST M_FaultCoolingTA  RST M_FaultLevelTA
RST   M_FaultHeatingTB  RST M_FaultCoolingTB  RST M_FaultLevelTB
RST   M_FaultHeatingTC  RST M_FaultCoolingTC  RST M_FaultLevelTC
MOV   K0 D_Suspicion
MOV   K0 D_AuxStage
MOV   K0 D_Ereq_kW
MOV   K0 D_NeedCount
MOV   K0 D_ServeCount
```

3.2 AI Channel Read/Scale

```
FROM  K0 K10 D0 K1
MUL   D0 K100 D2
DIV   D2 K4000 D4
```

3.3 Master Safety and Mode Control

```
LDI   X0
SET   M1_ESTOP
LD    X_EStopReset
RST   M1_ESTOP
LD    M1_ESTOP
OUT   Y0
LDI   M1_ESTOP
OUT   M2_SystemOK

LD    X1
OUT   Y1
LD    X1
OUT   M3_AutoEnabled
```

3.4 Manual/Automatic Gating Pattern (repeat per pump/valve)

```
LD    M2_SystemOK
ANI   M3_AutoEnabled
AND   X2
OUT   [pump output]

LD    M2_SystemOK
AND   M3_AutoEnabled
AND   [arbiter / need-array condition]
```


OUT [pump output]

3.5 Jacket Pump

```
LD M2_SystemOK
OUT Y4
```

3.6 PO Tank Designation

```
LD X_Confirm
PLS M100
LD M100
MOV D50 D51
```

```
LD= D51 K0 OUT M0
LD= D51 K1 OUT M1
LD= D51 K2 OUT M2
LD= D51 K3 OUT M3
```

3.7 10-Second Refresh Oscillator

```
LDI T0
OUT T0 K100
LD T0
OUT M10_Refresh
```

3.8 T_{water} Target

```
LD M10_Refresh
MUL D_Ptotal K300 D200
DIV D200 K6160 D202
ADD D202 K400 D204
```

```
LD< D204 K400
MOV K400 D204
LD> D204 K700
MOV K700 D204
MOV D204 D_Twater_target
```

3.9 Mode Register

```
LD= D_Mode K0 AND< D_AlphaS K85 MOV K1 D_Mode
LD= D_Mode K1 AND>= D_AlphaS K90 MOV K0 D_Mode
LD= D_Mode K1 AND< D_AlphaS K65 MOV K2 D_Mode
LD= D_Mode K2 AND>= D_AlphaS K70 MOV K1 D_Mode
LD= D_Mode K2 AND< D_AlphaS K45 MOV K3 D_Mode
LD= D_Mode K3 AND>= D_AlphaS K50 MOV K2 D_Mode
```

3.10 Suspicion Score

```
LD  M_Overrun_Pulse
INC  D_Suspicion
LD> D_Suspicion K3
MOV  K3 D_Suspicion
```

```
LD= D_Suspicion K1  OUT M_HMIPrompt
LD= D_Suspicion K2  OUT M_Alarm
LD= D_Suspicion K3  OUT M_DisableFeed
LD= D_Suspicion K3  OUT M_Alarm
```

```
LD  M_PromptActive
OUT T5  K3000
LD  T5
INC  D_Suspicion
RST  M_PromptActive
```

```
LD  M_DailyReset_Pulse
MOV  K0 D_Suspicion
```

3.11 E_{total} Calculation

Planning window = 5 minutes, proposed, not user-confirmed. Used to convert the loss terms below (Watts) into energy (Joules) over a forward-looking window, and again in Dynamic Power Evaluation (Section 3.13) to convert total remaining energy back into an average power figure. A shorter window (closer to the existing 10s master refresh) responds faster to real changes but is noisier; a much longer window (e.g. 1 hour, considered and rejected during development) risks holding a stale, too-high setpoint for longer than some real duty cycles (the PO tank's own dispense duty is ≈ 42 min) – 5 minutes was chosen as a middle point, not derived from any established figure elsewhere in this project.

Corrects a units error in the original E_5/E_C^* derivation: standing loss ($UA\Delta T$) is a *power* (W), but was previously added directly to a sensible-heat *energy* term (J) with no time conversion between them. Every loss term below is now multiplied by the planning window (K300, seconds) before being summed.

```
// E_alpha, tank TA -- only counts if TA is being served AND is a melt
// duty (not the PO tank). Piecewise per the base architecture: below
// 36C still needs sensible-to-melt + latent; 36-40C is liquid sensible
// only. Repeat identically for TB, TC.
LD M_Serve_TA
ANI M1
AND<= D_ThetaTA K360
SUB K360 D_ThetaTA D700          // (36.0 - ThetaTA), tenths degC
DMUL D700 K200 D702              // x Cp_solid(2000) -- sensible term
DADD D702 K158400000 D704        // + latent+loss allowance (scaled)
DMUL D704 D_MassTA D_EalphaTA   // x mass -> Joules

LD M_Serve_TA
ANI M1
AND> D_ThetaTA K360
AND<= D_ThetaTA K400
SUB K400 D_ThetaTA D710
DMUL D710 K210 D712              // Cp_liquid(2100)
DMUL D712 D_MassTA D_EalphaTA

// E_5 -- sensible term + loss term (now correctly window-scaled)
SUB D_TF D_ThetaS D720
DMUL D720 K210 D722
DMUL D722 D_Mass5 D_E5sensible   // Joules

SUB D_ThetaS K181 D724           // (ThetaS - ambient 18.1C)
DMUL K978 D724 D726              // U5*A5, scaled x1000 -- Watts x1000
DMUL D726 K300 D728              // x window(s) -> Joules x1000
DDIV D728 K1000 D_E5loss

DADD D_E5sensible D_E5loss D_E5
```

```

// E_C_star (PO tank) -- identical pattern to E_5, own U*A and mass
SUB  K300 D_ThetaC D730          // target 30.0C - ThetaC
DMUL D730 K210 D732
DMUL D732 D_MassPO D_EC_sensible

SUB  D_ThetaC K181 D734
DMUL K5908 D734 D736             // U_PO*A_PO, scaled x1000
DMUL D736 K300 D738
DDIV D738 K1000 D_EC_loss

DADD D_EC_sensible D_EC_loss D_EC_star

// Sum only branches actually being served this cycle
DADD D_EalphaTA D_EalphaTB D740
DADD D740          D_EalphaTC D742
DADD D742          D_E5          D744
DADD D744          D_EC_star    D_Etotal

```

Every intermediate above needs 32-bit (DMUL/DADD/DDIV) – same overflow risk as the T_f block, for the same reason (large constants, kg-scale masses). K978 (T5's UA , scaled) and K5908 (PO tank's UA , scaled) are carried over from the architecture document's established loss coefficients – confirm scaling factors match before use. D_ThetaC/D_MassPO assume the currently- designated PO tank's readings are copied into dedicated registers elsewhere (not shown) rather than re-resolving M1/M2/M3 inline here.

3.12 T_f Thermal Blending

```

// m_p: T5's remaining capacity to reach 100% full, from T5's own
// live mass -- NOT read from a T10 tank.
SUB  K4871 D_MassT5 D_Mp

MOV  D_ThetaH D330    MOV  K0 D331
MOV  D_ThetaS D332    MOV  K0 D333
MOV  D_Mp      D334    MOV  K0 D335

DMUL K75808 D330 D340
DMUL D334    K2000 D344
DMUL D344    D332 D350

DADD D340    D350 D360
DADD K75808 D344 D362

DDIV D360 D362 D370
MOV  D370 D_Tf

LD>= D_Tf K550

```

```

OUT  M_TfInterlock
LD   M_TfInterlock
OUT  M4_ThrottleOrDivert

```

K4871 is T5's full-tank mass at 2000 mm ($2.4353 \text{ kg/mm} \times 2000$), matching the height-to-mass constant established elsewhere in this document.

3.13 Dynamic Power Evaluation

```

// E_total -> average power over the planning window (Watts), then kW
LD  M10_Refresh
DDIV D_Etotal K300 D_PbaseW      // Joules / 300s = Watts
DDIV D_PbaseW K1000 D_Pbase      // Watts / 1000 = kW

// Condition 1: measured water temp < 40.0C (full power)
LD  M10_Refresh
AND< D_Twater K400
MOV D_Pbase D_Ereq_kW

// Condition 2: measured water temp >= 70.0C (zero power -- safety taper)
LD  M10_Refresh
AND>= D_Twater K700
MOV K0 D_Ereq_kW

// Condition 3: 40.0C <= measured water temp < 70.0C (proportional taper)
// -- unchanged from the original design; D_Twater here is MEASURED,
// not the target, so this is a protective ceiling-approach taper,
// independent of demand, layered on top of D_Pbase.
LD  M10_Refresh
AND>= D_Twater K400
AND< D_Twater K700
SUB K700 D_Twater D_TempDiff
MUL D_Pbase D_TempDiff D_TempCalc
DIV D_TempCalc K300 D_Ereq_kW

```

Fast override, proposed and only sketched for T5 – not a complete implementation. Purpose: prevent a stale (up to 5-min-old) high E_{req} from continuing to command power once a served tank has demonstrably already reached its own target, rather than waiting for the next planning-window refresh.

```
LD  M10_Refresh
AND< D_ThetaS D_Twater_target
OUT M_FastOverride_T5
```

```
LD  M_FastOverride_T5
MOV K0 D_Ereq_kW
```

This checks only T5. Extending to the PO tank and T10 melt duties (and combining via OR before the final MOV) is the same pattern but was not worked out per-duty here – treat this as a direction, not a finished block.

3.14 Auxiliary Stage Register

```
LD= D_AuxStage K0  AND> D_Ereq_kW K0  MOV K1 D_AuxStage
LD= D_AuxStage K1  AND<= D_Ereq_kW K2  MOV K0 D_AuxStage
LD= D_AuxStage K1  AND>= D_Ereq_kW K14 MOV K2 D_AuxStage
LD= D_AuxStage K2  AND< D_Ereq_kW K10  MOV K1 D_AuxStage
LD= D_AuxStage K2  AND>= D_Ereq_kW K26 MOV K3 D_AuxStage
LD= D_AuxStage K3  AND< D_Ereq_kW K22  MOV K2 D_AuxStage
```

```
// Now gated on M2_SystemOK -- was missing in the previous version
LD  M2_SystemOK
AND>= D_AuxStage K1
OUT Y7
```

```
LD  M2_SystemOK
AND>= D_AuxStage K2
OUT Y10
```

```
LD  M2_SystemOK
AND>= D_AuxStage K3
OUT Y20
```

3.15 Tank Scoring (worked for TA; repeat for TB, TC)

```
MUL D_ThetaTA K100 D410
DIV D410      K700 D_TnormTA
```

```
MUL D_AlphaTA K6 D412
MUL D_TnormTA K4 D414
ADD D412 D414 D_ScoreCasualTA
```

```
MUL D_AlphaTA K4 D416
MUL D_TnormTA K6 D418
```

```
ADD D416 D418 D_ScoreImmTA
```

3.16 PO Exclusion and Ranking

```
LD M1 MOV K-1 D_EligCasualTA
LDI M1 MOV D_ScoreCasualTA D_EligCasualTA
LD M2 MOV K-1 D_EligCasualTB
LDI M2 MOV D_ScoreCasualTB D_EligCasualTB
LD M3 MOV K-1 D_EligCasualTC
LDI M3 MOV D_ScoreCasualTC D_EligCasualTC

MOV K1 D_MaxID
MOV D_EligCasualTA D_MaxScore

LD> D_EligCasualTB D_MaxScore
OUT M_Challenger
LD M_Challenger MOV D_EligCasualTB D_MaxScore
LD M_Challenger MOV K2 D_MaxID

LD> D_EligCasualTC D_MaxScore
OUT M_Challenger
LD M_Challenger MOV D_EligCasualTC D_MaxScore
LD M_Challenger MOV K3 D_MaxID

LD= D_MaxID K1 MOV D_ThetaTA D_RankedTankTemp
LD= D_MaxID K2 MOV D_ThetaTB D_RankedTankTemp
LD= D_MaxID K3 MOV D_ThetaTC D_RankedTankTemp
```

3.17 Height-to-Mass Conversion

```
SUB K2000 D_DistT5 D_HeightT5
DMUL D_HeightT5 K24353 D510
DDIV D510 K10000 D_MassT5
// T10 tanks: SUB K2200 ..., DMUL ... K40455, same pattern
```

3.18 P1 – T5 Block

```
LD< D_ThetaS K400
OUT M_P1Demand

LD M_TfInterlock
OUT M4_ThrottleOrDivert

LD< D_AlphaS K50
AND>= D_RankedTankTemp K360
OUT M_BatchRelease
LD M_BatchRelease
OUT Y5
```

Y11 is intentionally *not* driven here – it is driven once, from M_Serve_T5, in the Need-Array Arbiter below. This resolves the double-drive issue flagged in the previous version of this document.

3.19 P2 – PO Tank Block

```
LD D_DispensedVol
AND>= D_DispensedVol D_2xNd
OUT M_Overrun_Pulse

LD M1 OR M2 OR M3
AND [dispense active, not yet "process complete"]
OUT M_P2Demand
```

3.20 Need-Array Arbiter

```
LD M_P1Demand          OUT M_D_T5
LD M1 AND M_P2Demand    OUT M_D_TA_asPO
LD M2 AND M_P2Demand    OUT M_D_TB_asPO
LD M3 AND M_P2Demand    OUT M_D_TC_asPO

LDI M1 AND< D_ThetaTA K360 OUT M_D_TA_asMelt
LDI M2 AND< D_ThetaTB K360 OUT M_D_TB_asMelt
LDI M3 AND< D_ThetaTC K360 OUT M_D_TC_asMelt

LD M_D_TA_asPO OR M_D_TA_asMelt OUT M_D_TA
LD M_D_TB_asPO OR M_D_TB_asMelt OUT M_D_TB
LD M_D_TC_asPO OR M_D_TC_asMelt OUT M_D_TC

MOV K0 D_NeedCount
LD M_D_T5 INC D_NeedCount
LD M_D_TA INC D_NeedCount
LD M_D_TB INC D_NeedCount
LD M_D_TC INC D_NeedCount

LD M_D_T5 OUT M_Serve_T5
LD M_D_TA OUT M_Serve_TA
LD M_D_TB OUT M_Serve_TB
LD M_D_TC OUT M_Serve_TC

LD= D_NeedCount K4 ANI M0 AND M1
LD< D_ScoreImmTB D_ScoreImmTC RST M_Serve_TB
LD= D_NeedCount K4 ANI M0 AND M1
LD>= D_ScoreImmTB D_ScoreImmTC RST M_Serve_TC

LD= D_NeedCount K4 ANI M0 AND M2
LD< D_ScoreImmTA D_ScoreImmTC RST M_Serve_TA
LD= D_NeedCount K4 ANI M0 AND M2
LD>= D_ScoreImmTA D_ScoreImmTC RST M_Serve_TC
```



```

LD= D_NeedCount K4 ANI M0 AND M3
LD< D_ScoreImmTA D_ScoreImmTB RST M_Serve_TA
LD= D_NeedCount K4 ANI M0 AND M3
LD>= D_ScoreImmTA D_ScoreImmTB RST M_Serve_TB

LD= D_NeedCount K4 AND M0
AND< D_ScoreImmTA D_ScoreImmTB AND< D_ScoreImmTA D_ScoreImmTC
RST M_Serve_TA
LD= D_NeedCount K4 AND M0
AND< D_ScoreImmTB D_ScoreImmTA AND< D_ScoreImmTB D_ScoreImmTC
RST M_Serve_TB
LD= D_NeedCount K4 AND M0
AND< D_ScoreImmTC D_ScoreImmTA AND< D_ScoreImmTC D_ScoreImmTB
RST M_Serve_TC

MOV K0 D_ServeCount
LD M_Serve_T5 INC D_ServeCount
LD M_Serve_TA INC D_ServeCount
LD M_Serve_TB INC D_ServeCount
LD M_Serve_TC INC D_ServeCount

LD M2_SystemOK AND M3_AutoEnabled AND= D_ServeCount K1 OUT Y6
LD M2_SystemOK AND M3_AutoEnabled AND= D_ServeCount K2 OUT Y2
LD M2_SystemOK AND M3_AutoEnabled AND= D_ServeCount K3 OUT Y17

LD M_Serve_T5 OUT Y11 LD M_Serve_TA OUT Y12
LD M_Serve_TB OUT Y13 LD M_Serve_TC OUT Y14

LD M2_SystemOK
ANI M3_AutoEnabled
AND X2
OUT Y2

```

3.21 Solar and Auxiliary Gating

```

LD M3_AutoEnabled
AND> D_ThetaSol D_Twater_target
OUT Y15

LDI M3_AutoEnabled
AND X10
OUT Y15

LD M3_AutoEnabled
AND<= D_ThetaSol D_Twater_target
OUT Y16

LDI M3_AutoEnabled

```

```
ANI X10
OUT Y16
```

```
LD Y15
OUT Y25
```

3.22 Oil Distribution

```
LD M_DemandX
OUT M_ServeX
```

```
LDI M_DemandX AND M_DemandY
OUT M_ServeY
```

```
LDI M_DemandX ANI M_DemandY AND M_DemandZ
OUT M_ServeZ
```

```
LD M_ServeX OUT Y22
LD M_ServeY OUT Y21
LD M_ServeY OUT Y24
LD M_ServeZ OUT Y21
LD M_ServeZ OUT Y23
```

```
LD M_ServeX OR M_ServeY OR M_ServeZ
OUT Y3
```

Open item carried forward: M_ServeX opens no T5 outlet valve (Y21) while M_ServeY/M_ServeZ both do – confirm whether this asymmetry is intentional before this goes to hardware.

3.23 Fault Detection

```
// --- TANK A ---
LD M10_Refresh
SUB D_ThetaTA D_ThetaTA_Prev D_DeltaThetaTA
LD M10_Refresh
MOV D_ThetaTA D_ThetaTA_Prev
```

```
DMUL K5011 K100 D600
DDIV D600 D_MassTA D602
DDIV D602 K2000 D_MaxHeatRateTA
```

```
SUB D_ThetaTA K181 D610
DMUL K668 D610 D620
DDIV D620 D_MassTA D622
DDIV D622 K2000 D_MaxCoolRateTA
```

```
LD> D_DeltaThetaTA D_MaxHeatRateTA
OUT M_FaultHeatingTA
```

```
LD< D_DeltaThetaTA KO
AND> D_DeltaThetaTA D_MaxCoolRateTA
OUT M_FaultCoolingTA

LD M10_Refresh
SUB D_MassTA D_MassTA_Prev D_DeltaMassTA
LD M10_Refresh
MOV D_MassTA D_MassTA_Prev

LD M_ServeTA_AODD
OR M_ServeX OR M_ServeY OR M_ServeZ
OUT M_TAActiveTransfer

LDI M_TAActiveTransfer
AND<> D_DeltaMassTA KO
OUT M_FaultLevelTA

LD> D_ThetaTA K750
OUT M_HighLimitTA
// (repeat identically for TB, TC)

LD M_HighLimitTA
OR M_HighLimitTB
OR M_HighLimitTC
SET M_Alarm

LD M_Alarm
RST Y7
RST Y10
RST Y20
```

3.24 Alarm Acknowledge and Reset

```
LD X_AlarmReset
ANI M_HighLimitTA
ANI M_HighLimitTB
ANI M_HighLimitTC
RST M_Alarm
```

4 Structured Text (ST) – EXPERIMENTAL

This section is experimental and unverified. GX Works2's ST support is primarily associated with Mitsubishi's iQ-R/iQ-F hardware family, not the classic FX-series line the ladder logic assumes – ST availability on the Coolmay L01S has not been established. Coverage is partial by design: straightforward blocks are omitted; every block with nontrivial arithmetic or branching is included.

4.1 E_{total} Calculation

```

IF M_Serve_TA AND (NOT M1) THEN
  IF D_ThetaTA <= 360 THEN
    D_EalphaTA := ((360 - D_ThetaTA) * 200 + 158400000) * D_MassTA;
  ELSIF D_ThetaTA <= 400 THEN
    D_EalphaTA := (400 - D_ThetaTA) * 210 * D_MassTA;
  END_IF;
END_IF;
(* repeat identically for TB, TC *)

D_E5sensible := (D_TF - D_ThetaS) * 210 * D_Mass5;
D_E5loss      := (978 * (D_ThetaS - 181) * 300) / 1000;
D_E5          := D_E5sensible + D_E5loss;

D_EC_sensible := (300 - D_ThetaC) * 210 * D_MassP0;
D_EC_loss     := (5908 * (D_ThetaC - 181) * 300) / 1000;
D_EC_star     := D_EC_sensible + D_EC_loss;

D_Etotal := D_EalphaTA + D_EalphaTB + D_EalphaTC + D_E5 + D_EC_star;

```

4.2 T_f Thermal Blending

```

D_Mp := 4871 - D_MassT5;    (* T5's own remaining capacity, not read
                             from any T10 tank *)

D_Tf := (75808 * D_ThetaH + D_Mp * 2000 * D_ThetaS) / (75808 + D_Mp * 2000);

M_TfInterlock := (D_Tf >= 550);
M4_ThrottleOrDivert := M_TfInterlock;

```

4.3 Dynamic Power Evaluation

```

IF M10_Refresh THEN
  D_PbaseW := D_Etotal / 300;    (* Joules / 5-min window = Watts *)
  D_Pbase  := D_PbaseW / 1000;   (* -> kW *)

  IF D_Twater < 400 THEN
    D_Ereq_kW := D_Pbase;
  ELSIF D_Twater >= 700 THEN
    D_Ereq_kW := 0;

```

```

ELSE
    D_TempDiff := 700 - D_Twater;
    D_Ereq_kW := (D_Pbase * D_TempDiff) / 300;
END_IF;
END_IF;

(* Fast override, T5 only -- proposed, not extended to other duties *)
M_FastOverride_T5 := M10_Refresh AND (D_ThetaS < D_Twater_target);
IF M_FastOverride_T5 THEN
    D_Ereq_kW := 0;
END_IF;

```

4.4 Auxiliary Stage Register

```

CASE D_AuxStage OF
    0: IF D_Ereq_kW > 0 THEN D_AuxStage := 1; END_IF;
    1: IF D_Ereq_kW <= 2 THEN D_AuxStage := 0;
        ELSIF D_Ereq_kW >= 14 THEN D_AuxStage := 2;
        END_IF;
    2: IF D_Ereq_kW < 10 THEN D_AuxStage := 1;
        ELSIF D_Ereq_kW >= 26 THEN D_AuxStage := 3;
        END_IF;
    3: IF D_Ereq_kW < 22 THEN D_AuxStage := 2; END_IF;
END_CASE;

Y7 := M2_SystemOK AND (D_AuxStage >= 1);
Y10 := M2_SystemOK AND (D_AuxStage >= 2);
Y20 := M2_SystemOK AND (D_AuxStage >= 3);

```

4.5 Tank Scoring and Ranking

```

D_TnormTA := (D_ThetaTA * 100) / 700;
D_TnormTB := (D_ThetaTB * 100) / 700;
D_TnormTC := (D_ThetaTC * 100) / 700;

D_ScoreCasualTA := (D_AlphaTA * 6) + (D_TnormTA * 4);
D_ScoreCasualTB := (D_AlphaTB * 6) + (D_TnormTB * 4);
D_ScoreCasualTC := (D_AlphaTC * 6) + (D_TnormTC * 4);

D_ScoreImmTA := (D_AlphaTA * 4) + (D_TnormTA * 6);
D_ScoreImmTB := (D_AlphaTB * 4) + (D_TnormTB * 6);
D_ScoreImmTC := (D_AlphaTC * 4) + (D_TnormTC * 6);

D_EligCasualTA := SEL(M1, D_ScoreCasualTA, -1);
D_EligCasualTB := SEL(M2, D_ScoreCasualTB, -1);
D_EligCasualTC := SEL(M3, D_ScoreCasualTC, -1);

D_MaxID := 1;
D_MaxScore := D_EligCasualTA;

```

```

IF D_EligCasualTB > D_MaxScore THEN
    D_MaxScore := D_EligCasualTB;
    D_MaxID := 2;
END_IF;
IF D_EligCasualTC > D_MaxScore THEN
    D_MaxScore := D_EligCasualTC;
    D_MaxID := 3;
END_IF;

CASE D_MaxID OF
    1: D_RankedTankTemp := D_ThetaTA;
    2: D_RankedTankTemp := D_ThetaTB;
    3: D_RankedTankTemp := D_ThetaTC;
END_CASE;

```

4.6 Need-Array Arbiter

```

M_D_T5 := M_P1Demand;
M_D_TA := (M1 AND M_P2Demand) OR (NOT M1 AND (D_ThetaTA < 360));
M_D_TB := (M2 AND M_P2Demand) OR (NOT M2 AND (D_ThetaTB < 360));
M_D_TC := (M3 AND M_P2Demand) OR (NOT M3 AND (D_ThetaTC < 360));

D_NeedCount := 0;
IF M_D_T5 THEN D_NeedCount := D_NeedCount + 1; END_IF;
IF M_D_TA THEN D_NeedCount := D_NeedCount + 1; END_IF;
IF M_D_TB THEN D_NeedCount := D_NeedCount + 1; END_IF;
IF M_D_TC THEN D_NeedCount := D_NeedCount + 1; END_IF;

M_Serve_T5 := M_D_T5;
M_Serve_TA := M_D_TA;
M_Serve_TB := M_D_TB;
M_Serve_TC := M_D_TC;

IF (D_NeedCount = 4) THEN
    IF M1 THEN
        IF D_ScoreImmTB < D_ScoreImmTC THEN M_Serve_TB := FALSE;
        ELSE M_Serve_TC := FALSE;
        END_IF;
    ELSIF M2 THEN
        IF D_ScoreImmTA < D_ScoreImmTC THEN M_Serve_TA := FALSE;
        ELSE M_Serve_TC := FALSE;
        END_IF;
    ELSIF M3 THEN
        IF D_ScoreImmTA < D_ScoreImmTB THEN M_Serve_TA := FALSE;
        ELSE M_Serve_TB := FALSE;
        END_IF;
    ELSIF M0 THEN
        IF (D_ScoreImmTA <= D_ScoreImmTB) AND (D_ScoreImmTA <= D_ScoreImmTC) THEN
            M_Serve_TA := FALSE;

```

```

        ELSIF (D_ScoreImmTB <= D_ScoreImmTA) AND (D_ScoreImmTB <= D_ScoreImmTC) THEN
            M_Serve_TB := FALSE;
        ELSE
            M_Serve_TC := FALSE;
        END_IF;
    END_IF;
END_IF;

D_ServeCount := 0;
IF M_Serve_T5 THEN D_ServeCount := D_ServeCount + 1; END_IF;
IF M_Serve_TA THEN D_ServeCount := D_ServeCount + 1; END_IF;
IF M_Serve_TB THEN D_ServeCount := D_ServeCount + 1; END_IF;
IF M_Serve_TC THEN D_ServeCount := D_ServeCount + 1; END_IF;

IF M2_SystemOK AND M3_AutoEnabled THEN
    Y6 := (D_ServeCount = 1);
    Y2 := (D_ServeCount = 2);
    Y17 := (D_ServeCount = 3);
END_IF;

Y11 := M_Serve_T5;
Y12 := M_Serve_TA;
Y13 := M_Serve_TB;
Y14 := M_Serve_TC;

```

4.7 Fault Detection

```

IF M10_Refresh THEN
    D_DeltaThetaTA := D_ThetaTA - D_ThetaTA_Prev;
    D_ThetaTA_Prev := D_ThetaTA;

    D_MaxHeatRateTA := (5011 * 100) / D_MassTA / 2000;
    D_MaxCoolRateTA := (668 * (D_ThetaTA - 181)) / D_MassTA / 2000;

    M_FaultHeatingTA := (D_DeltaThetaTA > D_MaxHeatRateTA);
    M_FaultCoolingTA := (D_DeltaThetaTA < 0) AND
        (ABS(D_DeltaThetaTA) > D_MaxCoolRateTA);

    D_DeltaMassTA := D_MassTA - D_MassTA_Prev;
    D_MassTA_Prev := D_MassTA;
END_IF;
(* repeat identically for TB, TC *)

M_TAAActiveTransfer := M_ServeTA_AODD OR M_ServeX OR M_ServeY OR M_ServeZ;
M_FaultLevelTA := (NOT M_TAAActiveTransfer) AND (D_DeltaMassTA <> 0);

M_HighLimitTA := (D_ThetaTA > 750);
M_HighLimitTB := (D_ThetaTB > 750);
M_HighLimitTC := (D_ThetaTC > 750);

```

```
IF M_HighLimitTA OR M_HighLimitTB OR M_HighLimitTC THEN
    M_Alarm := TRUE;
    Y7  := FALSE;
    Y10 := FALSE;
    Y20 := FALSE;
END_IF;
```


5 Conclusion

1. **E_{total} is now actually computed**, not just consumed – closing the largest gap in the previous version of this document. The units error in E_5/E_{C^*} 's loss terms (Watts added to Joules with no time conversion) is fixed by introducing the planning window explicitly.
2. **Two real, unconfirmed decisions were made to get here**: the 5-minute planning window (proposed over the 1-hour figure considered during development, on staleness grounds), and the T5-only fast override (a sketched direction, not a complete per-duty implementation). Both are flagged inline and in the Nomenclature/ID tables, not buried.
3. **D_{Mp} and the Y7/Y10/Y20 gating are both now resolved** – m_p is T5's own computed remaining capacity, and the auxiliary elements now correctly drop under E-STOP.
4. **The Condition-3 taper in Dynamic Power Evaluation is confirmed correct as originally written** – it operates on *measured* water temperature as an independent safety ceiling, not on the dynamic target, and needed no change.
5. Every other open item from the previous version (Y21 asymmetry in oil distribution, tie-break on equal scores, E-STOP reset source, mode power-up default) remains open and unchanged.

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Based on the *VCH Sizing Framework, Second Edition*
DOI: 10.5281/zenodo.21009246